⊗ 2021, 2015
Write the characteristic features of unimolecular reaction?

⇒ (I) The reaction involves only one reactant molecule. For example: A ⟶ Product

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(II) The reaction rate depends only on the concentration of the single reactant.

$$\text{Rate} = K[A]$$

where K is rate constant & [A] is the concentration of reactant.

(III) It follows first order kinetics.

(IV) Its rate ~~dermi~~ determining step involves only one molecule.

(V) It can be explained by Lindemann-Hinshelwood mechanism.

(VI) Unimolecular reaction often involve the formation of highly reactive intermediates.

(VII) Example: $N_2O_4(g) \longrightarrow 2NO_2(g)$.

⊗ 2018, 2015
Discuss Kinetic isotope effect. How does the kinetic isotope effect affect an equilibrium process? or, short note?

⇒ K. i. effect refers to the change in the rate of a chemical reaction due to the substitution of an atom with one of its isotopes. This effect is most commonly

observed when H is replaced with its havier isotopes, ^2H or ^3H. There are two types of Kinetic isotope effect.

① Primary Kinetic isotope effect is observed when the rate determining step requires the scission of a bond involving the isotope.

② secondary K.I. effect is the reduction in reaction rate even though the bond involving the isotope is not broken to form product.

KIE arises due to differences in the vibrational zero-poin energies of bonds involving lighter & ~~heav~~ heavier isotopes. A bond with lighter isotope has higher ZPE & easier to break, leading to faster reactions. A bond with heavy isotope has lower ZPE & requires more energy to break, leadin to slower reactions.

⇒ KIE can affect equilibrium process by altering (পরিবর্তন করা) the relative rates of the forward & reverse reaction. If the KIE affects one reaction more than the other, it can shift equilibrium position. If the forward reaction is

more sensitive to the KIE than the reverse reaction, the equilibrium will shift toward the reactants. If the reverse reaction is more sensitive the equilibrium will shift towards the products.

2017, 2016, 2015, 2020, 2018, 2021, 2022

(*) Describe the lindemann-Hinshelwood mechanism of unimolecular reactions? Mention its limitation? Significance at two ends?

⇒ It is a theoretical model that explains the kinetics of unimolecular reaction. In the lindemann-Hinshelwood mechanism it is supposed that a reactant molecule A becomes energetically excited by collision with another A molecule:

$$A + A \xrightarrow{K_a} A^* + A \qquad \frac{d[A^*]}{dt} = K_a[A]^2$$

The energized molecule (A^*) might lose its excess energy by collision with another molecule:

$$A + A^* \longrightarrow A + A \qquad \frac{d[A^*]}{dt} = -K_a'[A][A^*]$$

Alternatively, the excited molecule might shake itself apart & from product P. That is, it might undergo the unimolecular decay:

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$A^* \longrightarrow P \qquad \frac{d[A^*]}{dt} = -K_b[A^*]$

Apply the steady-state approximation, the net rate of formation of $[A^*]$,

$$\frac{d[A^*]}{dt} = K_a[A]^2 - K_a'[A][A^*] - K_b[A^*]$$

$$\Rightarrow [A^*] = \frac{K_a[A]^2}{K_b + K_a'[A]}$$

So the rate formation of P is,

$$\frac{d[P]}{dt} = K_b[A^*] = \frac{K_bK_a[A]^2}{K_b + K_a'[A]}$$

case-I: At sufficiently high pressure, $K_a'[A] \gg K_b$ then ~~eqn~~ rate becomes:

$$\text{rate} = \frac{K_bK_a[A]^2}{K_a'[A]} = K[A] \quad \left[\text{where, } K = \frac{K_bK_a}{K_a'}\right]$$

$\Rightarrow$ rate = constant x [A]

$\therefore$ rate $\propto$ [A]

This is first order reaction, independent of pressure.

case-II: At low pressure $K_b \gg K_a'[A]$ then the rate law,

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$$\text{rate} = \frac{K_b K_a [A]^2}{K_b} = K_a [A]^2$$

$\therefore$ rate $\propto [A]^2$

This is second order reaction.

Now the theory can be tested by defining an effective rate constant K; which would be constant if the reaction is first order at all process. Therefore,

$$K = \frac{K_b K_a [A]}{K_b + K_a' [A]}$$

$$\Rightarrow \frac{1}{K} = \frac{1}{K_a [A]} + \frac{K_a'}{K_b K_a}$$ where K = ~~effective~~ effective rate constant

If we plot $1/K$ vs $1/[A]$ then we will find a straight line with stope $\frac{1}{K_a}$.

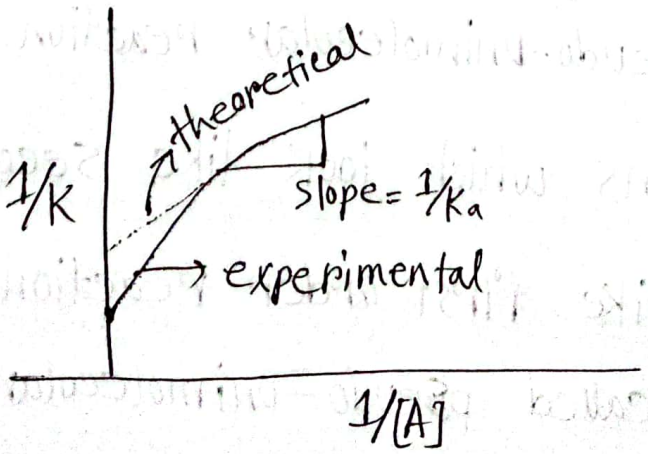


Fig: Lindemann-Hinshelwood plot for unimolecular reaction.

limitations: (I) This model fails to explain isotope effects or quantum tunneling in unimolecular reactions. (II) The mechanism assumes a simple activation-deactivation proce neglecting the complexities of energy redistribution within molecules. (III) The model assumes all collisions are equally effective at activating A, which is not true (IV) This mechanism is primarily applicable to gas phase reactions. It does not consider solvent effects in liquid-phase reactions (V) It predicts rate, ~~is independent~~ at high pressure, is independent of pressure. but experimental evidence often shows that the rate can still increase slightly with pressure, even at high pressure.

2022, 2020, 2018

(8) Define pseudo-unimolecular reaction with eqn & example

⇒ The reactions which look like second order reaction but behaves like first order reaction & follow firs order rate law are called pseudo-unimolecular reaction. This happens when one of the reactant is present in a much larger concentration compared to the others, making

its concentration effectively constant during the reaction.

equation: For a bimolecular reaction:

$$A + B \longrightarrow \text{product}$$

The rate law is: $\text{Rate} = K[A][B]$

If $[B] \gg [A]$ then $[B]$ remains nearly constant. then the rate law, $\text{Rate} = K'[A]$; where $K' = K[B]$

Example: acid-catalyzed hydrolysis of an ester.

$$CH_3COOC_2H_5 + \underset{(\text{excess})}{H_2O} \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

The rate law is: $\text{Rate} = K[CH_3COOC_2H_5][H_2O]$

since water is in large excess, its concentration remains nearly constant. So rate law becomes:

$$\text{Rate} = K'[CH_3COOC_2H_5]$$

other examples are hydrolysis of sucrose, Inversion of can sugar, hydrolysis of acetyl chloride etc.

⊗ 2015 what is unimolecular reaction?

⇒ It is a chemical reaction where a single reactant

molecule changes into products. It is a chemical r. in which the rate determining step involves only one reactant molecule. The r. rate depends only on the concentra of the single reactant. Rate = K[A]

Example : $N_2O_4(g) \longrightarrow 2NO_2(g)$

2015
Write the names of the methods for determining rate constant of a fast reaction ?

⇒ Methods : ① flow method ② stopped flow method ③ flash Photolysis method ④ laser induced-fluorescence method. ⑤ Resonance flurescence method. ⑥ shock tubes meth

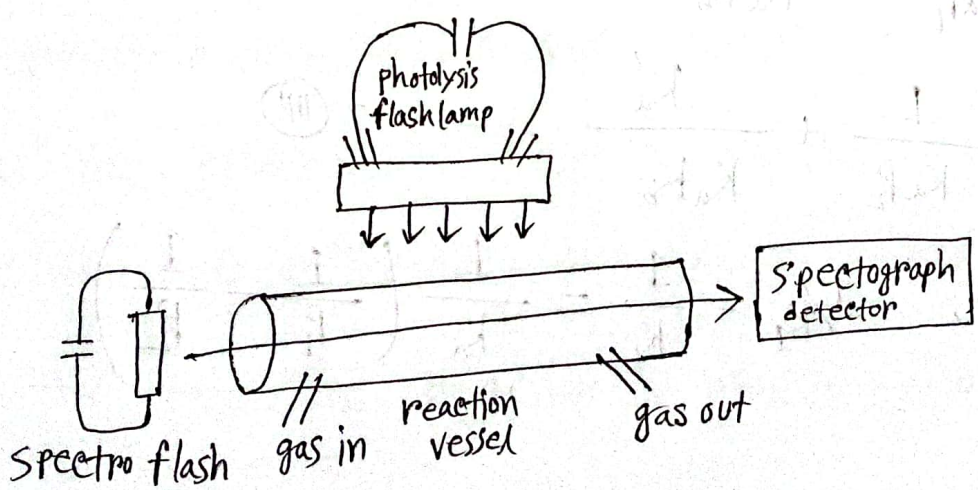
2015
Discuss the principle of flash photolysis technique ?

⇒ very fast reaction can be studied by flash photolysis, in which the sample is exposed to a ~~big flash~~ brief flash of light that initiates the reaction & then the contents of the reaction chamber are monitored. A high intensity, short duration flash of light is used to

excite molecules in the sample. The flash provides energy to break chemical bonds, excite molecules to higher energy states, or initiate photochemical reactions. This excitation can lead to the formation of short lived species like radicals, ions or excited states. Lower-intensity light source is used to monitor the absorption or emission of light by the transient species as a function of time. The light in the form of a laser excites a molecule into a reactive state usually in the form of a free radical. Reactions occuring on a nanosecond (10^{-9}s) may be monitored by flash photolysis method using electronic absorption or emission, IR absorption or raman scattering. An example of radicale generation:

$$Cl_2 + h\nu \longrightarrow Cl\cdot + Cl\cdot$$

$$Cl\cdot + H_2 \longrightarrow HCl + H\cdot$$



2016

The effective rate constant that has lindemann-Hinshelwood mechanism is $2.5 \times 10^{-4} s^{-1}$ at 1.30 kPa & $2.10 \times 10^{-5} s^{-1}$ at 12 kP calculate the rate constant for the activation step in the mechanism?

⇒ let us consider the following unimolecular reaction for Lindemann-Hinshelwood mechanism :-

$$A + A \underset{k'_a}{\overset{k_a}{\rightleftharpoons}} A^* + A \xrightarrow{k_2} \text{products}$$

for this reaction, the effective rate constant is :-

$$K = \frac{k_a k_b [A]}{k_b + k'_a [A]} = \frac{k_a k_b P}{k_b + k'_a P} \quad [\text{consider } [A] = P]$$

$$\Rightarrow 1/K = \frac{1}{k_a P} + \frac{k'_a}{k_a k_b} \quad \text{---- (I)}$$

For two different pressure eqn (I) become,

$$\frac{1}{K_1} = \frac{1}{k_a P_1} + \frac{k'_a}{k_a k_b} \quad \text{---- (II)}$$

$$1/K_2 = \frac{1}{k_a P_2} + \frac{k'_a}{k_a k_b} \quad \text{---- (III)}$$

Now, (II) − (III), $1/K_1 - \frac{1}{K_2} = \frac{1}{k_a}\left(\frac{1}{P_1} - \frac{1}{P_2}\right)$

$$\Rightarrow K_a = \frac{1/P_1 - 1/P_2}{1/K_1 - 1/K_2}$$

$$\Rightarrow K_a = \frac{\left(\frac{1}{1.30} - \frac{1}{12}\right) \times 10^{-3}}{\left(\frac{1}{2.5\times10^{-9}} - \frac{1}{2.10\times10^{-5}}\right)}$$

Here,
$K_1 = 2.5 \times 10^{-9}\ s^{-1}$
$K_2 = 2.10 \times 10^{-5}\ s^{-1}$
$P_1 = 1.3 \times 10^{+3}\ Pa$
$P_2 = 12 \times 10^{3}\ Pa$
$K_a = ?$

$\therefore K_a = -1.572 \times 10^{-8}\ Pa^{-1}s^{-1}$ Ans

2022, 2021

(✱) write short note on stopped flow system?

⇒ The stopped flow method is based on rapidly mixing two or more reactants & then observing the resulting reaction in real time.

Two sln containing the reactants are rapidly driven from syringes into a high-efficiency mixture. This mixer ensures the sln are throughly combined in a matter of microseconds. The mixed sln is then rapidly pushed into an observation cell (usually a cuvette). A stop syringe triggers the "stopped" part of the process. It stops the flow of the sln into the observation cell, effectively trapping the mixed reactants within the cell. The reaction is monitored within the observation

cell using a detection system, often a spectrometer. This system measures changes in absorbance or fluorescence as the reaction progresses.

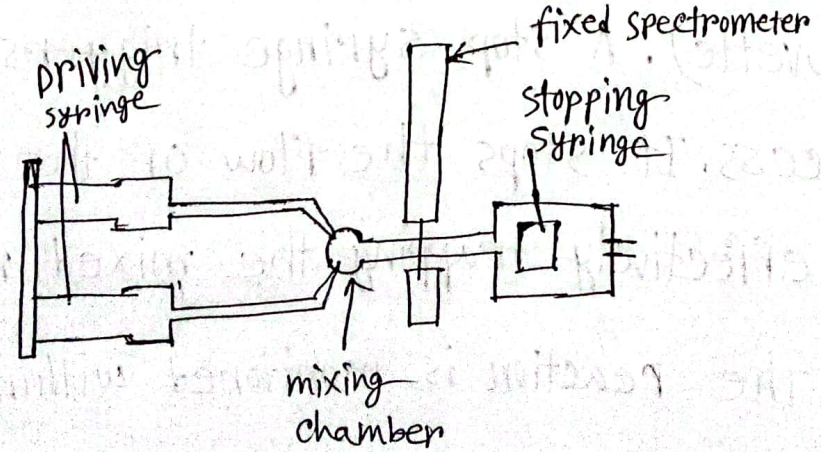
Stopped flow system is widely used in various field such as analysing drug interactions & stability, investing reaction mechanisms, studing enzyme kinetics, & ligand bonding etc.

Advantages: (i) This method is widely used for study of the protein folding & enzyme kinetics & ligand bonding.

(ii) This method requires small amount of sample.

(iii) this method allows for the study of reaction on the miliseco to second scale.

Disadvantages: (i) This method can be more complex to opera compared to other method. (ii) It is excellent for milisecond to second, but it is not ideal for slower reaction.



2017, 2016
Distinguish btwn pre-equilibrium approximation & steady-state approximation?

⇒

pre-equilibrium approximation	steady-state approximation
The rate of formation of product follows 2nd order rate law.	The rate of formation of product follows first order rate law.
It requires the first step to be faster than the second step	It requires 1st step of a reaction is much slower than second step
Assumes that the reactant and intermediate are in equilibrium.	Assumes that the concentration of the intermediate remains constant. because its rate formation equals its rate of consumption.
The step following the equilibrium is the slowest & rate determining	the rate determining step can be any step in the mechanism.
EX. $A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} I \xrightarrow{k_2} P$	EX. $A \xrightarrow{k_1} I \xrightarrow{k_2} P$
It requires a well defined equilibrium constant for the first step	May not be applicable if the reaction Mechanism is complex
It assumes that the reactant & intermediates of a multi step reaction exist in dynamic equilibrium	It assumes that the rate of change of conc. of all reaction intermediates are negligibly small. $\frac{d(\text{intermediate})}{dt} \approx 0$

2017
What is elementary reaction?

⇒ It is a chemical reaction in which one or more chemical species react directly to form products in a single step

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reaction & with a single transition state. EX:

$$H + Br_2 \longrightarrow HBr + Br$$

(★) 2017 write the differential eqn for the following consecu elementary reaction: $A \xrightarrow{K_a} I \xrightarrow{K_b} P$?

⇒ let us consider, the reaction,

$$A \xrightarrow{K_a} I \xrightarrow{K_b} P$$

The rate of unimolecular decomposition of A is,

$$\frac{d[A]}{dt} = -K_a[A] \quad \text{---} -(I)$$

The rate of formation of intermediate,

$$\frac{d[I]}{dt} = K_a[A] - K_b[I] \quad \text{---} (II)$$

for The product, $\frac{d[P]}{dt} = K_b[I]$ ---(III)

the rate of ordinary first order decay becomes,

$$[A] = [A_0]\, e^{-K_a t} \quad \text{---} (IV)$$

put this valve in eqn (II) we get,

$$\frac{d[I]}{dt} + K_b[I] = K_a[A_0]e^{-K_at} \quad \text{---(V)}$$

The differential eqn has a standard form & after setting $[I]=0$ the sln is,

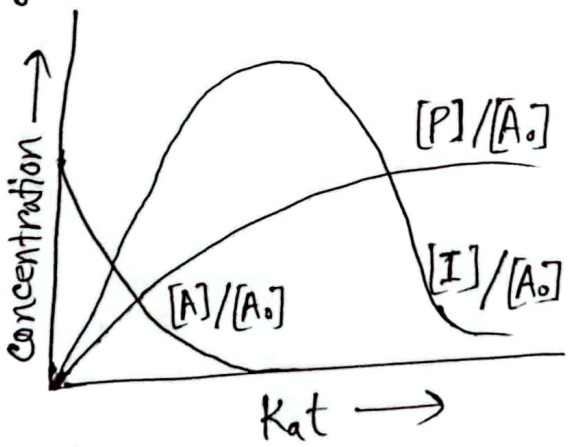
$$[I] = \frac{K_a}{K_b - K_a} \times \left(e^{-K_at} - e^{-K_bt}\right)[A_0] \quad \text{---(VI)}$$

at all times, $[A] + [I] + [P] = [A_0]$

so it follows that,

$$[P] = \left(1 + \frac{K_a e^{-K_bt} - K_b e^{-K_at}}{K_b - K_a}\right)[A_0] \quad \text{---(VII)}$$

This is the differential eqn for the above consecutive elementary reaction.



2020, 2018, 2017

(X) State steady-state approximation & write short note about it ?

⇒ State: This principle state that when short lived reaction

intermediates exist at low concentration in a system the rate of formation of the intermediates can be considered to be equal to their rate of disappearance.

It is a technique used in chemical kinetics to analyze complex reaction mechanisms with multiple steps. It assumes that the conc. of certain reactive intermediates remains relatively constant during the reaction.

$$\frac{d[\text{intermediate}]}{dt} \approx 0$$

consider a reaction, $A \xrightarrow{K_a} I \xrightarrow{K_b} P$

we know, $\frac{d[I]}{dt} \approx 0$ — (I)

Now, formation of I is, $\frac{d[I]}{dt} = K_a[A] - K_b[I]$

$$\Rightarrow K_a[A] - K_b[I] = 0$$

$$\Rightarrow [I] = \frac{K_a}{K_b}[A] \quad \text{— (II)}$$

The rate of formation of P is, $\frac{d[P]}{dt} = K_b[I]$

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$$\Rightarrow \frac{d[P]}{dt} = K_b \times \frac{K_a}{K_b}[A]$$

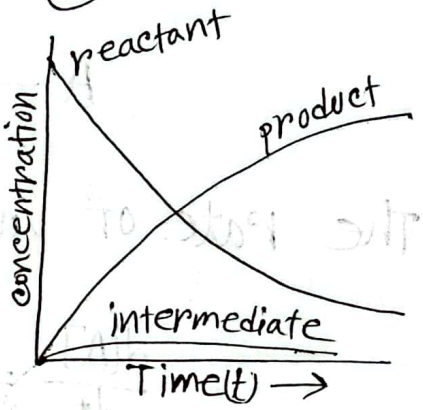
$$\therefore \frac{d[P]}{dt} = K_a[A] \quad \text{——(III)}$$

Initially the conc. of A is $[A_0]$, then $[A] = [A_0]e^{-K_a t}$

Then from eqn (III), $\frac{d[P]}{dt} = K_a[A_0]e^{-K_a t}$ ——(IV)

Then integrating eqn (IV),

$$[P] = [A_0](1 - e^{-K_a t})$$



The steady state a. is a valuable technique for simplifying complex reaction mechanisms & deriving rate laws.

2020, 2018, 2021

(X) Define consecutive reaction? Write short note on consecutive reaction?

⇒ A consecutive elementary reaction is a multi-step reaction where the product of one step becomes the reactant for the next step. These reactions often involve intermediate species that are formed & consumed during process. the general example is: $A \xrightarrow{K_a} B \xrightarrow{K_b} C$

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where A = initial reactant, B = reactive intermediate and C = final product, $K_b >> K_a$ so the first step is rate determining step.

Ex. $^{239}U \xrightarrow{23.5\ min} {}^{239}NP \xrightarrow{2.35\ min} {}^{239}Pu$.

⇒ let us consider, an consecutive elementary reaction,

$$A \xrightarrow{K_a} I \xrightarrow{K_b} P$$

The rate of unimolecular decomposition of A is,

$$\frac{d[A]}{dt} = -K_a[A] \quad \text{——(I)}$$

Now, $\frac{d[I]}{dt} = K_a[A] - K_b[I]$ ——(II)

&, $\frac{d[P]}{dt} = K_b[I]$ ——(III)

we know, $[A] = [A_0]\, e^{-K_a t}$ ——(IV)

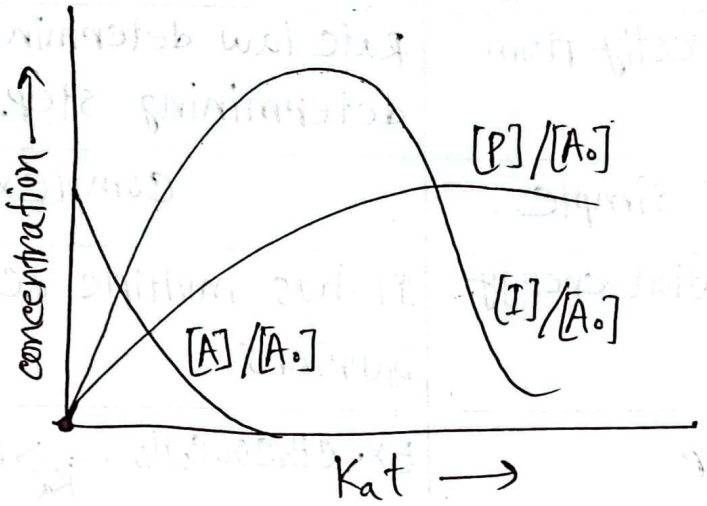
Now, from (II), $\frac{d[I]}{dt} + K_b[I] = K_a[A_0]\, e^{-K_a t}$ ——(V)

the differential eqn has a standard form & after setting $[I] = 0$, the sln is, $[I] = \frac{K_a}{K_b - K_a}\left(e^{-K_a t} - e^{-K_b t}\right)[A_0]$ ——(VI)

at all times, $[A] + [I] + [P] = [A_0]$

so, $$[P] = \left(1 + \frac{K_a e^{-K_b t} - K_b e^{-K_a t}}{K_b - K_a}\right)[A_0] \quad \text{(VII)}$$

This is the differential eqn for the above consecutive reaction.



2020

⊛ Distinguish btwn elementary & consecutive elementary reaction?

elementary	consecutive elementary
occurs in single step	occurs in multiple step
No intermediates	Intermediate are formed & consumed
Rate law determine Directly from molecularity.	Rate law determined by the rate-determining step.
overall reaction is simple.	complex
It has a single activation energy barrier	It has multiple activation energy barriers
EX. $H + Br_2 \longrightarrow HBr + Br$	EX. $CH_3COOC_2H_5 \xrightarrow[K_a]{} CH_3COOH \xrightarrow[K_b]{} CH_3COONa$

2021

⊗ Differentiate btwn primary & secondary Kinetic isotope effect?

primary	secondary
· less sensitive	more sensitive
Its KIE value is comperatively high	comperatively small
It is affected by non-vibrational factors	affected by vibrational factors
Describes the isotopic substitution at the broken bond.	Describes ..... at the adjacent bond to the broken bond.
PKIE is the change of r. rate due to isotopic substitution at the sight of bond breaking	SKIE is ..... substitution at a site other than the bond breaking site.
EX. Deuterium substitution at a carbon involved in a C-H bond cleavage.	EX. Deuterium substitution at a carbon adjacent to the reaction center.

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